

Managing Your Project

Using Software Engineering Process

CPT202 Software Engineering Group Project 2024/2025

Soon Phei Tin

Topics

- Exploring the projects in this module
- Understanding Scrum in action
- Get ready for the Cardboard game

Exploring 3 Projects

- To be available

Initial Requirements Study

- Understand Your Project -

- Problem statement
 - A problem statement is a concise description of an issue to be addressed or a condition to be improved upon.
 - It identifies the gap between the current state and desired state of a process or product.
 - The process of defining the problem is often a group effort. It starts with meeting with the stakeholders, customers, and/or users affected by the issue (if possible) and learning about their pain points.

Aims of the project

- What the project is set to achieve? What problems to resolve?
- Time frame?
- Phases?

Project Scope

- What does and doesn't lie within the project
 - E.g. The project track the status of order delivery. However, the actual delivery of the order is to be done by third party and hence is outside of the control of the system.
- Represent the scope using use-case diagram

User Characteristics

- List the users involved in the system.
- If they are organizational users, get the organizational chart.
- Understand the political influences of the users.
- List the characteristics of the end users that may influence UI/UX design.
 - Regional and cultural background
 - Age and ability in the system operation

Constraints

- Compatibility with existing internal/external system?
- Availability of the technologies?
- Environmental (e.g. pandemic)?
- Financial budget?
- Time constraint?

Assumption and Dependencies

- **Assumption:**
 - things that are accepted as true or as certain to happen, without proof.
 - the events or conditions that are most likely to occur when a project life-cycle takes place.
- Examples:
 - All the resources (equipment, tools, or materials) will be in good condition to be used throughout the project life-cycle.
 - The total cost of daily operations will remain unchanged.
 - All relevant stakeholders will come to the next meeting as scheduled.
 - All the physical devices such as laptops, computers, mobile phones, printers, and more are in good condition to be used when conducting the project.

Assumption and Dependencies

- **Dependencies:**
 - A dependency in project refers to a series of tasks that are interrelated.
 - Types of dependency
 - **Finish-to-Start:** Task B cannot start until task A has been completed
 - **Start-to-Start:** Task B cannot start until task A starts
 - **Finish-to-Finish:** Task A cannot be completed until task B is completed
 - **Start-to-Finish:** Task B cannot be completed until task A starts
- **Examples:**
 - There is an old website running live, but you want to replace the old one with the new one, you will retain the old website unless the new one is completed.
 - The configuration of server cannot be started before the completion of a training.

Research Task

- How to manage and reduce risks of your assumption and dependencies?

Scrum Framework

Software Requirements

- Epics > Features > Product Backlog Items > Tasks
- Epic is a larger scope of requirement which can be divided into smaller requirements
- An Epic can be spread across sprints and even across agile teams
- An Epic can be a high-level description of what the client wants
- Epics also allow teams to give high-level understanding of their work to stakeholders without getting into technical details.

Epic

- Examples:

- As a customer, I want to shop online so that I can purchase books from the online book store.
- As a store manager, I want to manage the store's overall operation so that the store operation is smooth and the customer is satisfying with the service.
- As an administrator, I want to maintain the book store website so that the store is secure and reliable.

- All epics must be in a SLAP

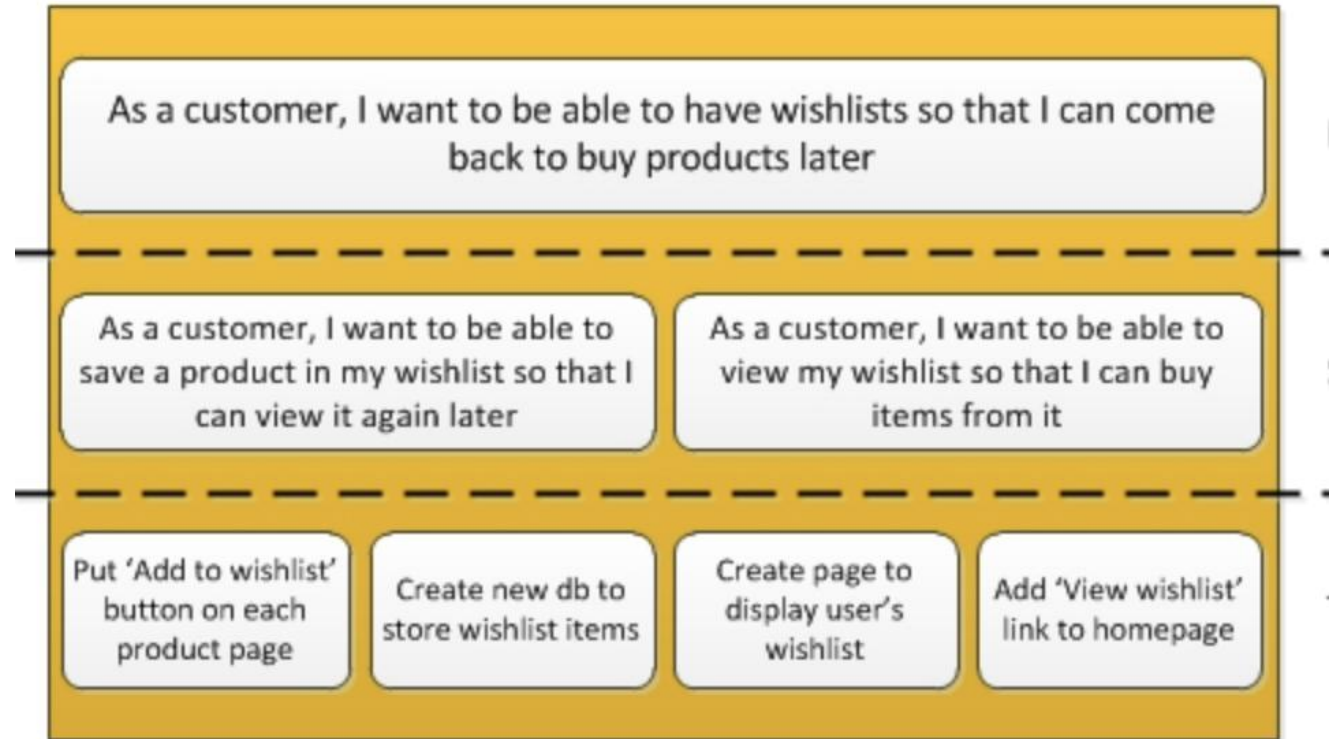
- Single Level of Abstraction Principles

Epic >> Features

- Epic: As a customer, I want to shop online so that I can purchase books from the online book store.
- Features:
 - As a customer, I want to have wish lists so that I can come back to buy products later.
 - As a customer, I want to check my order status so that I know the delivery progress of my order.
 - etc

Feature > PBI > Tasks

- **Feature** - A requirement that the business wants. It represents one of the features broken down from an Epic.
- **Product Backlog Item** - A detail requirement that we further defined from the Feature. It is something that is deliverable within a single sprint.
- **Tasks** - Steps need to be taken in order to fulfill the requirement stated in the PBI.



Sample Software Requirements

- E-Commerce Platform:
 - This is a large project and hence you may want to define multiple epics.
 - The PBIs were ignore for simplicity.
- To-Do List project:
 - It is a small project, we do not need to define epic.

Product Backlog

What is a Product Backlog?

- Product Owner is the one responsible to create all the Epics, Features, and PBIs.
- They are representing software requirements at different level of abstraction.
- They are all organized and stored as a collection we called Product Backlog.
- In this project, for educational purposes, we do not require the Product Owner. All team members are responsible to work collectively.

What is a PBI?

- A PBI is consisting of: -
 - PBI ID
 - Priority
 - Effort
 - Title
 - Story
 - Acceptance Criteria
 - Grooming Information
 - Some organization includes the following: -
 - Discussion
 - Related PBI

How Big a PBI Should Be?

- Refers to To-Do List project.
- Small enough to be completed within a Sprint.
- A Sprint usually consisting of multiple PBIs.
- Small and simple to understand and implements.
- Consisting of single purpose.

Features and PBIs – Bad Examples

- As a customer, I want to be able to visit the home page, so that I can see the website information.
- As an engineer, I want to design a database table, so that I can add a new book information to the database.
- As manager, I want to select some or all of the order contents and click the blue "pick up" button at the top right of the page to cancel the order and send the items to the user to complete the checkout. Managers can also filter orders by pickup location using the drop down menu on the left side of the top. Managers can view all pickup records. The request record will be removed from the list after the pickup.

Acceptance Criteria

- Refers to To-Do List project.
- *Acceptance Criteria* are a formal list that fully enumerates user requirements and all the product scenarios put into the account.
- It states the intent of the client and not the solution
- describe achievable and sensible information
- provide the minimum level of functionality the product is to achieve

Acceptance Criteria

- Example
 - User story: As a registered user I want to see the membership expiry date on the billing page so that I know when I have to manually renew or enable auto-renew.
 - Acceptance Criteria:
 - The membership expiry date is calculated.
 - The membership expiry date is displayed.
 - The date format is yyyy-mm-dd.
 - Membership details (including expiry date) are not displayed if the user is not registered/authenticated.
 - Formatted: GIVEN the registered user is authenticated WHEN visiting the billing page THEN the system calculate and display expiry date in yyyy-mm-dd format.

Given-When-Then

- (Given) some context
- (When) some action is carried out
- (Then) a particular set of observable consequences should obtain

An example:

- Given my bank account is in credit, and I made no withdrawals recently,
- When I attempt to withdraw an amount less than my card's limit,
- Then the withdrawal should be complete without errors or warnings

Sample Acceptance Criteria in Azure Board

We can include images to make the PBI clearer

In fact, it is a good practice to include images to help understanding.

In your assignment, try to use images appropriately.

PRODUCT BACKLOG ITEM 78*

78 As a registered user I want to see the membership expiry date on the billir

Unassigned

0 comments [Add tag](#)

State ● New

Area CPT202

Reason 🔒 New backlog item

Iteration CPT202\Sprint 1

Description

Use this area during PBI grooming session. See some examples below.

Can the member renew the membership manually before the expiry date?

What should be done if the member renew the membership manually?

Where on the billing page should I display the expiry date?

Is there any specific UI requirements?

How to calculate the expiry date?

Acceptance Criteria

Scenario: Authenticated user visiting billing page

GIVEN the registered user is authenticated WHEN visiting the billing page THEN the system calculate and display the expiry date in yyyy-mm-dd format.

Scenario: Unauthenticated user visiting billing page

GIVEN the registered user is not authenticated WHEN visiting the billing page THEN the system route the user to the login page.

Sample Acceptance Criteria

As a logged-out user

I want to be able to sign in to a website

So that I can find access my personal profile

Scenario: System user signs in with valid credentials

“Given I’m a logged-out system user

and I’m on the Sign-In page

When I fill in the “Username” and “Password” fields with my authentication credentials

and I click the Sign-In button

Then the system signs me in”

More Complex Acceptance Criteria

- More complex acceptance criteria can be achieved using AND clause
- Given-When-Then-AND-When-Then.....
- Suitable for actions with consequences for a common context

Sample Acceptance Criteria

As a website user

I want to be able to search on the webpage

So that I can find necessary information

According to the Given/When/Then template, the acceptance criteria would be the following:

Scenario: User searches for an item by its name

“Given that I’m in a role of registered or guest user

When I open the “Products” page

Then the system shows me the list of all products

And the system shows the “Search” section in the right top corner of the screen

When I fill in the “Search” field with the name of existing item in the product list

And I click the “Apply” button OR press the Enter key on keyboard

Then the system shows products in the Search Results section with product names matching entered product name

And the system shows the number of search results in the top of the Search Results section”

Important for your project

- Your team do not need a Product Owner. All members work collectively as a “Virtual” Product Owner.
- When working as Product Owner, you need to set your perspective as one.
- As requested in Coursework 1, each member must develop 5 PBIs.

PBI Grooming

- PBI Grooming is a process to refine the PBI so that it: -
 - Covers as much important information as possible.
 - Is in the right size.
- During grooming, we also estimate the effort required to implement it. (You are not required to estimate the effort of your PBI in this project)
- Grooming is a team effort.

For Coursework 1, grooming is not required!!!

PBI Grooming

- Covers as much important information as possible
 - Team members raise questions to gain more information that was missed out by the Product Owner.
 - Raise as many questions as possible to gain enough understanding so that you know what is required to develop.
 - The Product Owner is responsible to answer all the questions.
 - The questions and answers has to be recorded as part of the PBIs for future references.
- Is in the right size

PBI Grooming

- When do you groom?
 - Before starting Sprint 1: You need to groom enough PBIs with high priority.
 - During Sprint: Groom enough PBIs for the future Sprint to continue.
 - DO NOT groom too many PBIs for the Sprints in far future.

Sprint

What is a Sprint?

- A Sprint is a time box
 - Usually 2 weeks. Begin on first workday of week 1, and end on the last workday of week 2.
 - It may also be a 3-week or a 4-week Sprint.
 - Consistent throughout the project
 - Within the 2-week time box, you will do the following: -
 - Sprint Planning
 - Sprint Execution
 - Sprint Review
 - Sprint Retrospective

What is a Sprint?

- Work arrangement for a Sprint in sequence
 - Begin with a Sprint Planning.
 - It is a meeting among all team members to produce a Sprint Backlog
 - It takes less than 3 hours, depending on the experience of the developer and the difficulty of the project.
 - Follow by Sprint Execution. All team members begin the implementation following the Sprint Backlog.
 - Sprint Execution start from the 1st day of a Sprint after the Sprint Planning Meeting. Ends at the last day of the Sprint.
 - Throughout the Sprint Execution, conduct a Daily Scrum (Stand up meeting. It takes about 15 minutes)
 - Participant in PBI grooming whenever is needed.
 - Sprint Review
 - Sprint Retrospective

Sprint Planning

What is Sprint Planning?

- Sprint Planning is a meeting among all team members to produce a Sprint Backlog
- In practice, the Product Owner will tell the team the overall goal for the current Sprint. For your project, you should discuss and agree among all members. Or the team leader should fix an overall goal.
- According to the goal, select the PBIs to be implemented in this current Sprint.
- For each PBI to be implemented, discuss and agree on the tasks involve in completing the PBI.
 - For each task, who will do it? How much time is required?
- Record all information on a Sprint Backlog (Refer to the template)

Sprint Review & Sprint Retrospective

Sprint Review

- In practice, Sprint Review is a meeting among the development team and the stakeholders to review the result (software product) of the current Sprint.
- For your project, you have no actual stakeholders. Therefore, you can review the result among yourselves.
- It takes less than 2 hours, depending the on project. In your case, you can complete it within an hour or less.
- The meeting start on the last day of the Sprint just before Sprint Retrospective.
- The main objective of the review is to check how much the result fit the requirement of the stakeholders.
- If not fit, you may propose to redo the PBI in the future Sprint.

Sprint Retrospective

- Sprint Retrospective is a meeting among development team members to review the process of the current Sprint.
- Members can comment and suggest the current Sprint in the spirit of making the process better.
- Finalize all the comments and suggestions and decide three most important suggestions for enhancement in the next Sprint.
- The meeting should not run for more than 2 hours. You could control to make it less than 1 hour.
- It is the last thing to do in a Sprint.

Scrum Cardboard Game

- Refer to the attached Word document